

Share My UL — Discounted Ticket

Architecture, Cloud Platforms & Implementation Reference

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Tech Stack: Next.js + Supabase + Vercel

Running Cost: \$0.00 / month forever

1. Executive Summary & Core Concept

Share My UL is an automated, web-based ticket sharing application designed for students in Uppsala. It allows students in a private WhatsApp group to view real-time availability of an unused UL transit season ticket, select exact 12-hour or 24-hour time slots, and submit instant booking requests over WhatsApp without needing to register user accounts.

💡 Why It Was Built

To monetize and share season ticket downtime with student friends at affordable rates (25 SEK for 12h, 30 SEK for 24h) while automating scheduling, timezone math, and awake handover constraints.

🚀 100% Free Stack (\$0/month)

Leveraged permanent free tiers: Vercel Edge Serverless Hosting, Supabase Managed PostgreSQL, Google Cloud OAuth 2.0, and GitHub Cloud Repositories.

2. Step-by-Step Platforms Breakdown & Setup Actions

● Google Cloud Console — Authentication & OAuth Setup

Purpose: Provides secure, passwordless one-tap admin authentication for Vikas via Google Sign-In.

Actions Performed:

- **OAuth Consent Screen:** Created an External app named *Share My UL*, set support & developer email to `vikasvaradhana@gmail.com`, and published to Production mode.
- **OAuth 2.0 Client Credentials:** Created Web Application Client ID and Secret.
- **Authorized JavaScript Origins:** Added `http://localhost:3000` and `https://share-my-ul.vercel.app`.
- **Authorized Redirect URIs:** Pointed to Supabase callback: `https://pjdedywxobkfupuumdxf.supabase.co/auth/v1/callback`.

● Supabase — Managed PostgreSQL & Auth Provider

Purpose: Hosts the relational database, executes atomic schedule queries, and handles OAuth session tokens.

Actions Performed:

- **Database Schema & Seeds:** Created `settings`, `schedule_blocks`, and `reservations` tables with Row Level Security (RLS).
- **Auth Providers:** Enabled Google OAuth and supplied the Google Client ID & Client Secret.
- **URL Configuration:** Configured **Site URL** to `https://share-my-ul.vercel.app` and added **Redirect URLs** (`https://share-my-ul.vercel.app/**` and `http://localhost:3000/**`).
- **API Security:** Extracted Project URL, Public Anon Key, and Secret Service Role Key for backend administration.

● GitHub — Source Control & Deployment Trigger

Purpose: Cloud repository hosting the codebase and triggering automated continuous deployment to Vercel.

Actions Performed:

- **Git Setup & Security:** Initialized git repository with strict `.gitignore` rules protecting `.env.local` secret keys.
- **Remote Repository:** Linked and pushed to `https://github.com/vikasvaradhana-lab/share-my-ul.git` on branch `main`.

▲ Vercel — Cloud Hosting & Serverless Execution

Purpose: Builds and hosts the Next.js full-stack app, executes API route handlers, and distributes traffic over a global CDN.

Actions Performed:

- **Project Import:** Imported `share-my-ul` repository from GitHub with 1-click CI/CD integration.
- **Environment Variables Configured:** Injected 5 production environment variables (Supabase URL, Anon Key, Service Role Key, WhatsApp Number, Admin Email).
- **Live Production Deployment:** Assigned live global URL at `https://share-my-ul.vercel.app` with free automatic SSL.

💬 WhatsApp Integration — Direct Ticket Booking Link

Purpose: Generates frictionless deep links directly to Vikas's WhatsApp with complete booking details.

Actions Performed:

- **Number Sanitization:** Formats admin number `+46793366464` to `https://wa.me/46793366464`.
- **Custom Message Template:** Pre-fills: "Hi there, I'd like to request the UL ticket. Date: [date] Time: [start] → [end] Duration: [12h/24h] Price: [SEK]".

3. Key Business Rules & Timezone Engine

- **Three-Timezone Architecture:**
 - *Storage:* UTC ISO timestamps in database.
 - *Display:* Stockholm local time (`Europe/Stockholm`) with automatic summer/winter daylight saving time.
 - *Admin Awake Constraints:* Verified against Indian Standard Time (`Asia/Kolkata` 06:30–22:30 IST) for both start and end handover moments.
- **Interactive Slider:** 30-minute increment range slider with live duration, price, and handover validation.
- **Cross-Midnight Window:** Merges contiguous available slots across consecutive green days for 24h slots.
- **Timeline Color Rules:**
 - `Available` : Active bookable periods.
 - `Reserved for me` : Personal travel days/slots.
 - `Reserved` : Booked student slots.
 - `Past / Unbookable` : Elapsed earlier hours of today or dates past booking cutoff.

4. Admin Portal Features (`/admin`)

📅 Month Filter & Accordion

Month tabs (`[All Months]` , `[August · Now]` , `[September]`).
Current day (Today) is open by default; all other days collapsed into neat 1-line summary rows.

⚙️ Personal Time Management

Dedicated `♥ Reserve for Me` button. Configurable recurring Wednesday & Friday hours (e.g. 08:30–17:30 or 24h), editable directly from the schedule list.

5. Environment Variables & Quick Reference

Variable Name	Value / Description
<code>NEXT_PUBLIC_SUPABASE_URL</code>	<code>https://pjdedywxobkfupuumdxf.supabase.co</code>
<code>NEXT_PUBLIC_SUPABASE_ANON_KEY</code>	Public anonymous Supabase JWT key
<code>SUPABASE_SERVICE_ROLE_KEY</code>	Admin service role secret key (never exposed to client)
<code>WHATSAPP_NUMBER</code>	<code>+46793366464</code>
<code>ADMIN_GOOGLE_EMAIL</code>	<code>vikasvaradhana@gmail.com</code>

Resource	Location
Live App	https://share-my-ul.vercel.app/
Admin Login	https://share-my-ul.vercel.app/admin
GitHub Repo	https://github.com/vikasvaradhana-lab/share-my-ul
Local App Copy	E:\1. My AG Apps\share-my-ul

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